

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the working of MRI with a neat and labeled block diagram.
- Q.24 Draw the block diagram of CT and explain its system components.
- Q.25 Explain ultra sound Imaging in detail and write its five properties.

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4th Sem / Medical Electronics

Subject : Medical Imaging Techniques

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which one is not a diagnostic equipment
- a) Multi para monitor b) MRI
- c) CT d) USG
- Q.2 USG uses
- a) Magnetic effect b) Doppler effect
- c) Photo Effect d) None
- Q.3 What does “MRI” stand for
- a) Magneto-Ray Idometry
- b) Medical Radiometry Instrument

- c) Magnetic Resonance Imaging
- d) Maximal Radiology Imaging

Q.4 Line focus principle is used in

- a) USG b) Xray
- c) MRI d) ECG

Q.5 Stationary anode type x ray tube is used in

- a) Xray b) Ventilator
- c) Defibrillator d) USG

Q.6 Which of the following sound frequencies would include diagnostic ultrasound?

- a) 10 Hz b) 10KHz
- c) 100 kHz d) 10 MHz

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write one use of a gamma camera.

Q.8 Write name of one EM wave.

Q.9 Write the definition of angular momentum.

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Q.10 Write one biological effect of USG.

Q.11 Write one safety requirement for X-ray.

Q.12 Stationary anode type X ray tube is used in_____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write a short note on A mode in USG.

Q.14 Explain the principle of fluoroscopy in a short note.

Q.15 Write a short note on tube rating.

Q.16 Write a short note on NMR coils.

Q.17 Draw the block diagram of X ray Machine.

Q.18 Write a short note on coherent scattering.

Q.19 Explain the angiography procedure in a short note.

Q.20 Describe the Biological effects of Ionizing radiation in a short note.

Q.21 Write a short note on permanent magnet MRI.

Q.22 Explain line focus principle in a short note.

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